

Curriculum Vitae / Resume

Avinash Gyawali

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SUMMARY

Second year master's student in computer vision with a research focus on efficient ML, Video MLLMs, 3D reconstruction and streaming video understanding.

EDUCATION

University of Central Florida (UCF)	2025 - 2027
<i>M.Sc. in Computer Vision</i>	<i>GPA: 3.8/4.0</i>
New York University Abu Dhabi	2021 - 2025
<i>Bachelor of Computer Science</i>	<i>GPA: 3.78/4.0</i>

PUBLICATIONS

[Under Review] StreamPrune: Causal and Query-Free Token Pruning in Streaming Video MLLMs	
*Avinash Gyawali, , Rajat Modi , Yushang Zhang, Yogesh S Rawat	May 2026
[PRE-PRINT] PatientCare: A Benchmark for Fine-Grained Clinical Action Understanding in Patient Safety	
*Avinash Gyawali, Priyank Pathak, Sirshapan Mitra, Shruti Vyas, Yogesh S Rawat	May 2026
[PRE-PRINT] Segmented Visual Geometry Grounded Transformer	
Avinash Gyawali, Yushra Ahmed	August 2025
[ACCEPTED] A Custom Haptic Syringe to Improve Virtual Reality Local Anesthesia Surgical Training	
Avinash Gyawali, Mohamad Eid, Marci Levine, Elizabeth C McAlpin	November 2024

EXPERIENCE

Center for Research in Computer Vision , UCF	Orlando , USA
<i>Computer Vision Research Assistant</i>	<i>August 2025 – Nov 2025</i>
- Designed and implemented a knowledge distillation pipeline to transfer zero-shot image classification capabilities from CLIP's ViT-based visual encoder (e.g., FastViT) to a lightweight ResNet-50 backbone. - Developing a pre-encoder token-pruning method for video MLLMs — run-length tokenization that merges temporally redundant patches across frames — to run long-video inference under a fixed compute budget with no loss in accuracy. - Developing streaming video MLLMs that combine token compression and semantic-aware pruning to sustain high-FPS perception under fixed compute budgets, targeting latency-critical settings such as autonomous driving and robotics.	
Applied Multimedia Lab, NYUAD	Abu Dhabi, UAE
<i>Research Assistant</i>	<i>August 2024 – May 2025</i>
- Developed a emotion classification model and optimized a transformer-based ATCNet model to classify five emotional states from EEG signals, improving model accuracy from an initial 68% to a peak of 74.45% - Engineered a deep learning pipeline for emotion recognition by implementing an ATCNet architecture that uses convolutional layers, multi-head self-attention, and temporal convolutional networks to effectively extract spatiotemporal features from raw EEG data	
Empathic Computing Lab	Adelaide, Australia
<i>HCI Research Intern, Advised by Dr. Mark Billinghurst</i>	<i>June 2024 – December 2024</i>
- Fine-tuned and deployed a Llama-8B LLM using parameter-efficient QLoRA to accurately classify deictic language from live speech, enabling real-time, low-latency visual cues in a collaborative VR application. - Engineered an end-to-end machine learning pipeline in Unity that integrated OpenAI's Whisper for speech-to-text with the custom-tuned LLM to seamlessly translate conversational intent into visual aids.	

AWARDS

NYUAD Summer Fellowship: Received grant to pursue research in Australia(\$5000)
Microsoft Hololens Hackathon 🏆: 1st place among 10 universities in the UAE (\$8000 Prize Money) 📄
Meta AWS WebXR Hackathon 🏆: 2nd place among 15 teams 📄
Microsoft Imagine Cup 🏆: Top 10 finalists from Asia (Project Name: EDVR)
Tie Startup Competition 🏆: Top 10 finalists from the Middle East
NASA Space Apps Regional 🏆: Top 2 among 20+ teams

SKILLS

Programming languages: Python, C++, C#, SQL
ML Tools: PyTorch, OpenMMLab , HuggingFace Transformers, CLIP, OpenCV, LoRA , Jupyter, Pandas, NumPy , DINO
Cloud Frameworks: AWS , Azure ML
Human Languages: English, Nepali